

Cryptocurrency Investment Product Data Design and Analysis Report

Event-Driven Rebalancing Based on Sentiment Spikes

Station #1: Data Collection and Processing

a) Product Description

This project delivers *TILT.ai*, a cryptocurrency portfolio optimisation service that rebalances holdings in response to abnormal sentiment shifts detected in crypto-related news. The system uses high predictive correlation to its advantage: the concept behind how sentiment in news and media can significantly influence future returns. It integrates structured market data (price and volume) with unstructured data (news headlines) to detect spikes or drops in sentiment. These events trigger reallocation of portfolio weights, enabling users to react dynamically to market mood. The goal is to enhance risk-adjusted returns by blending price trends with narrative signals, helping users avoid overexposure during fear periods or capitalise on positive sentiment rallies.

b) Product Basic Use and Interface

The product is designed as a hybrid system. It will support historical analysis and backtesting (offline mode), and real-time insights using API feeds (online mode).

A GET request to the CoinDesk API will be used to pull both price and news data periodically. The current implementation sets this as daily, but with only a few modifications (indicated by comments in the code) it can be configured to support hourly querying. While web sockets may provide superior real time data fetching, the current implementation only necessitates daily (or hourly) querying. Thus a GET request is simpler, more stable and sufficient for the use case.

The system will use historical Open High Low Close Volume (OHLCV) and news data to evaluate strategy performance. It focuses on event-driven sentiment signals rather than traditional factors like momentum or reversal, however, users may still choose to overlay simple technical filters.

The system will scan for daily (or hourly) sentiment shifts and perform rebalancing on a weekly basis, but this can be triggered earlier when sentiment anomalies are detected.

Interaction with the system will be via a simple web dashboard or notebook environment (e.g. Streamlit or Jupyter), showing the current sentiment state, recommended portfolio weights and backtest results.

c) Input Data

Structured Data will be retrieved from CryptoCompare, accessed via the CoinDesk API. It will get the top 50 crypto assets (BTC, ETH, and select alt coins), and for each coin retrieve the date time and historical OHLCV data. It will retrieve the data daily (or hourly) to align to news time resolution.

Unstructured Data will be retrieved from the same CoinDesk API. It will retrieve all news articles from the API over a time period set by the user when they run the code (default set to past 3 months). It will get the date of when the article was published, the headline, the text body, and other useful information beyond the scope of this part of the report, such as keywords list and initial sentiment (which is simple 0 or 1 (negative or positive) score) provided by the API.

d) Data Processing Requirements

Processing structured data will involve calculating daily returns and rolling volatility, aligning price data across selected crypto assets, and handle missing or duplicate time entries. This is quite a straightforward process, as the majority of structured data processing will be influenced in Station 2 feature engineering.

Processing unstructured data will be much more involved. This will be challenging, because a lot of retrieved articles contain spam, while other articles may still hold crucial market-moving information, even if they do not specify a specific crypto coin or token. Filtering out irrelevant articles early will ensure Station 2 will have a much more straightforward text processing pipeline.

Firstly, irrelevant columns will be dropped to preserve space in the data frame. Then, the title and body fields are combined into one "content" field, as the sentiment analysis in Station 3 will require analysis of all text in a single column. All text in the content field will be covered to lowercase, to make it easier to compare text without needing to deal with case. If an article contains no text in this field, it is dropped.

Then articles containing spam text will also be dropped using a technique known as masking. This is because, after inspection of some of the retrieved articles, most spam articles do not contain any meaningful news. A curated list of spam phrases was used during this filtering process. This was the most simple option, and more robust NLP practices may be used in the future to achieve this same filtering more reliably.

Finally, all articles will be assigned the symbol of the coin that appears the most prevalent in the article. If an article does not mention any specific coin, but it discusses relevant market-moving macroeconomic events, it will be given the "MARKET" symbol. Again, a curated list of keywords will be used, as it will be a simple yet effective way of incorporating broader economic sentiment. All other articles without any sort of symbol are then deemed irrelevant and thus dropped.

This rigorous process of processing unstructured data will be necessary for preparation of basic text cleaning, applying sentiment scoring, sentiment score aggregation and detection of

anomalies in Station 3. Market articles may be targeted and analysed by sentiment separately to determine whether to even optimise portfolio weights in the first place.

Lastly, all historical and news data will be stored in separate CSV files for now. In the future, lightweight databases such as SQLite for structured data, and MongoDB for unstructured document-based data, could be used for better data integrity. Both datasets will be aligned by timestamp for feature fusion.

e) Output Data

The final cleaned dataset will include, per day and per asset:

- date
- price return, rolling volatility, volume
- news article text, keywords and initial sentiment positive or negative score

This dataset will feed directly into the rebalancing logic and backtest engine.

f) Station #1 Definition

Station 1 establishes the foundation of the data factory floor, by collecting, cleaning and aligning structured crypto historical OHLCV data with unstructured sentiment signals from news articles. From a data management perspective, temporal alignment, data integrity and feature readiness are ensured for the downstream portfolio optimisation engine. This station takes raw market signals and human sentiment and derives actionable, clean datasets for decision making and backtesting in preparation for the next station.

Station #2: Feature Engineering

a) Input Requirements

To engineer effective features for a cryptocurrency portfolio strategy, the following inputs are required

- Clean OHLCV price data for each crypto data. This will come from the structured historical OHLCV dataset in station 1, which holds open, high, low and close prices, and traded volume
- Return series, derived from closing prices, including daily/weekly returns and log returns
- News articles and metadata. This will come from the unstructured news dataset in station 1, which includes date of publication, tags and coin mentions
- Specific rolling periods for computing statistics such as volatility. For example, 7 day, 14 day or 28 day windows
- Benchmark index, which is a market-wide index. For example, a total crypto market cap or Bitcoin as proxy, to compute excess returns or for normalisation

b) Data Collection and Format Requirements

To ensure quality and usability of these features, both structured and unstructured data needs to be aligned by timestamp (date time column in UTC). Daily resolution is preferred. Since crypto markets operate 24/7, the system assumes continuous time, thus UTC is used as standard. This

alignment of news data to market data will ensure intra-day resolution, because timestamps will be bucketed into day-level aggregates.

The following file organisation will be used

```
data
|- results
| |- clean_data
| | |- <SYMBOL>_ohlcv.csv
| | |- stage_1_news_raw.csv
| | |- stage_2_crypto_data.csv
| | |- stage_2_common_words.csv
| | |- stage_2_news_clean.csv
```

Unified naming across all stages (stations) will ensure traceability.

c) Core Features – Structured Data

From crypto price and volume data:

1. Return Features

- Daily Return:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

- Log Return:

$$r_t^{log} = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

- Momentum (k-day cumulative return):

$$M_t^{(k)} = \frac{P_t}{P_{t-k}} - 1$$

- Usefulness: Captures trends and investor behaviour.

2. Risk Features

- Rolling Volatility (σ):

$$\sigma_t^{(k)} = \sqrt{\frac{1}{k-1} \sum_{i=t-k+1}^t (r_i - \bar{r})^2}$$

- Value-at-Risk (VaR): Using historical or parametric methods to estimate 5% percentile of returns.
- Maximum Drawdown: Measures worst peak-to-trough loss in a rolling window.

3. Liquidity and Volume Metrics

- Daily Volume
- Volume Z-Score
- Rolling Average Volume
- All features will be normalised or ranked as needed for factor modelling.

d) Text Processing Pipeline

The text processing pipeline follows 5 steps, 4 for this station alone.

Step 1: Basic Text Cleaning

re, BeautifulSoup, str.lower() and UTF-8 encoding libraries and techniques will be used to ensure clean and uniform base text before NLP. Challenges include removing embedded HTML or non-UTF characters. Some metrics to consider will be the percentage of malformed or dropped articles.

Step 2: Text Normalisation

The nltk library will be used for tokenisation and lemmatisation, and stop word removal using built-in libraries will ensure trivial words like "and" and "be" are removed. Despite these not included in stop word removal, a crypto-specific dictionary may be used in the future for specialised terms. Challenges include management of crypto jargon and ticker-symbol ambiguity. Some metrics to consider will be token length distribution and vocabulary size.

Step 3: Content Filtering

Currently the CoinDesk API returns useful keywords contained in the article, and curated lists were used in Station 1 to filter out irrelevant articles. In the future, this should be upgraded to use dedicated content filtering techniques to identify relevant articles. This can be achieved through named entity recognitions for keyword extraction, fuzzy matching to filter irrelevant articles, and duplicate detection using hash techniques. Challenges include articles that mention multiple unrelated tokens. Some metrics to consider include retention rate after filtering, and rate of duplicate entries.

Step 4: Feature Extraction

Features include article length, content sentiment and frequency per token. Temporal patterns include moving average of article count, for example, BTC daily article count as a proxy for attention. Some metrics to consider include text feature variance and daily article counts.

Step 5: Data Integration *(Reserved for Station 3)*

Finally, cleaned news features should be joined with market data on date or timestamp. Thus a combined dataset with both text-derived and market-derived variables would be returned. Challenges include sparse news days or lag in sentiment effects. Some metrics to consider will be merge success rate and percentage of missing values.

This has been migrated to Station 3 (Model Design), because this station would invoke extracting the news features, aggregating per day and per symbol, and joining the features with OHLCV data. This will prevent temporal leakage, and lots of duplicate price rows and unusable news rows, due to the complexity of some news articles being general market news not specific to a single crypto.

e) Back-and-Forth Between Station #1 and Station #2

Feature engineering may expose anomalies not visible in the raw format. For example, outliers in volume or missing price data. Some articles may be irrelevant or falsely labeled as crypto-related, which would necessitate improved filtering or better keyword sets. Rolling features may require continuous data, so gaps from station 1 may need imputation or interpolation. And the text pipeline may surface encoding errors or misaligned timestamps that prompt a re-pull or raw news.

f) Station #2 Definition

Station 2, from a data management perspective, is the transformation layer between raw collection and predictive modelling. It extracts structured features from historical OHLCV data, and unstructured insights from news articles, aligned by a common time base. These features represent the inputs for factor modelling and portfolio optimisation of weights, balancing quantitative price and volume signals with qualitative sentiment and relevance metrics. This station ensures dual-source feature integrity for reliable, robust investment signal generation.